Luyao Zhong

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EDUCATION

Hong Kong University of Science and Technology (HKUST) Feb. 2026 – Present
Ph.D. in Physics

Chongqing University Sep. 2021 – Jun. 2025
B.Sc. in Physics

RESEARCH EXPERIENCE

Theoretical & Computational Study of Unconventional Superconductivity 2025 – 2026
Ph.D. student *HKUST*

- Developed symmetry-based classifications of superconducting pairing channels in multiorbital quantum systems using group representation theory.
- Constructed effective Hamiltonians and BdG models and analyzed their symmetry and topological properties using analytical and numerical methods.
- Worked extensively with operator representations, basis transformations, low-energy projections, and symmetry constraints in complex quantum systems.

First-Principles Calculation of Electronic and Thermal Transport 2023 – 2025
Undergraduate student *Chongqing University*

- Performed first-principles and Boltzmann-transport calculations of electronic and lattice thermal transport in topological and quantum materials.
- Analyzed the interplay among electronic structure, electron–hole compensation, and thermal/electrical transport.

PUBLICATIONS

[1] Upward deviation from the Wiedemann-Franz law and phonon-dominated thermal transport in the topological semimetal NbP_2
Jie Yang, **Luyao Zhong**, Xiaolong Yang. *Physical Review B* 114, 204302 (2026).
DOI: https://doi.org/10.1103/c1h3-z72r

[2] Large violation of the Wiedemann–Franz law driven by electron-hole compensation in the topological semimetal CoSi
Luyao Zhong, Xin Jin, Mingquan He, Rui Wang, Xiaoyuan Zhou, Tianqi Deng, Xiaolong Yang. *Physical Review B*, 113, L121102 (2026).
DOI: https://doi.org/10.1103/5svq-g1k7

[3] Electrical and thermal transport properties of the kagome metals ATi_3Bi_5 (A= Rb, Cs)
Xintong Chen, Xiangqi Liu, Wei Xia, Xinrun Mi, **Luyao Zhong**, Kunya Yang, Long Zhang, Yuhan Gan, Yan Liu, Guiwen Wang, Aifeng Wang, Yisheng Chai, Junying Shen, Xiaolong Yang, Yanfeng Guo, Mingquan He. *Physical Review B*, 107(17), 174510 (2023).
DOI: https://doi.org/10.1103/PhysRevB.107.174510

TECHNICAL BACKGROUND

- **Theoretical Physics:** symmetry and group representation theory, quantum mechanics, statistical mechanics, and condensed-matter theory.
- **Computational Methods:** Python, numerical linear algebra, and high-performance scientific computing.

HONORS & AWARDS

- **Top-10 Poster Award**, Quantum Materials Symposium, HKUST(GZ) 2025
- **Yan Jici Scholarship**, Institute of Physics, CAS 2023
- **Outstanding Student Scholarship**, Chongqing University 2022 – 2024